Grep

Definition

- Grep i used to search text in a giving file(s). Grep works line by line.(it maches the search criteria in a line by line basis)

Usage/formula

- `grep` + `option` = `search criteria` + `files`
- ## Examples
  - Find word ignoring case
    - `grep -i "mo" users.txt`
  - Search any line that contains the world "dracula" in the giving file:
    - `grep 'dracula' ~/Documents/dracula.txt`
  - Search any line that contain any word that contain "dracula' regardless of the case.
    - `grep -i 'dracula' ~/Documents/Books/dracula.txt`
  - Display how many lines contain the mach string
    - `grep -c 'dracula' ~/Documents/Books/dracula.txt`
  - Search any line that contain the word dracula regardless of case with number line
    - `grep -in 'dracula' ~/Documents/Books/dracula.txt`
  - Search all the lines that do not contain the word 'war'
    - `grep -v 'war' ~/Documents/Books/war-and-peace.txt`
  - Search and display only the matching string(pattern)
    - `grep -o 'pride' ~/Documents/Books/war-and-peace.txt`
  - Display a list of users with the /bin/bash/ login shell
    - `grep -i "bin/bash" /etc/passwd`
  - Display the user's information as store in the /etc/passwd
    - `grep -i $USER /etc/passwd`

Awk

Definition

- Awk is a scripting language used for processing and displaying text. Awk can work with a text file or from the standard output.

Usage/formula

- `akw` + `options` + `{awk command}` +`file` + `file to save (optional)`

Examples

- Print the first column of every line
  - `awk '{print $1}' ~/Documents/Csv/cars.csv`
- Print the last field of /etc/passwd file
  - `awk -F '{print #NF}' /etc/passwd`
- Print the first and the last field of the /ect/passwd
  - `awk -F '{print $1, " = " , $NF}' /etc/passwd`
- Print the first and 3 field of the /ect/passwd
  - `awk -F '{print NR,$1,$3}' /etc/passwd`
- Print the first and the 4th field with a different field separator
  - `awk -F '{OFS="="}{print $1,$4}' /etc/passwd`
- Start printing a file from a giving line(exclude the first 2 lines)
  - `awk 'NR > 3 {print}' /etc/passwd`
- Convert the first field to upper/lower case
  - `awk -F: '{print toupper($1)}' /etc/passwd`
- Print the length of a line(record)
  - `awk '{print length($0)}' /etc/passwd`
- Print specific field based on a command output. For example, the size and file name
  - `ls -lhF documents/ | awk 'BEGIN {printf "%s\t%s\n",:"Size","Name"} {print $5, "\t,$9}`

Sed

Definition

- SED is a string editor that perform operations on files and standard output. For instant it can search and replace,insert,and deleted. By using SED you can edit files without opening them

Usage/formula

- `sed` + `options` + `sed script` + `file`

Examples

- Replacing a string in givin file globally(replace false for true)
    - `sed 's/false/true/g' ~/Documents/sample_files/Json.json`
- Replacing only the fourth occurrence per line in a file
    - `sed 's/false/true/4' joke.json`
- Replacing from the giving number occurrence to the rest occurrence in a file. Start at the second time the word appears and continue to till end of the file
    - `sed 's/false/true/3g' joke.json`
- Replacing string on a specific line number
    - `sed '3 s/false/true/' joke.json`
- Replace string on a range of lines
    - `sed '1,3 s/false/true/' joke.json`